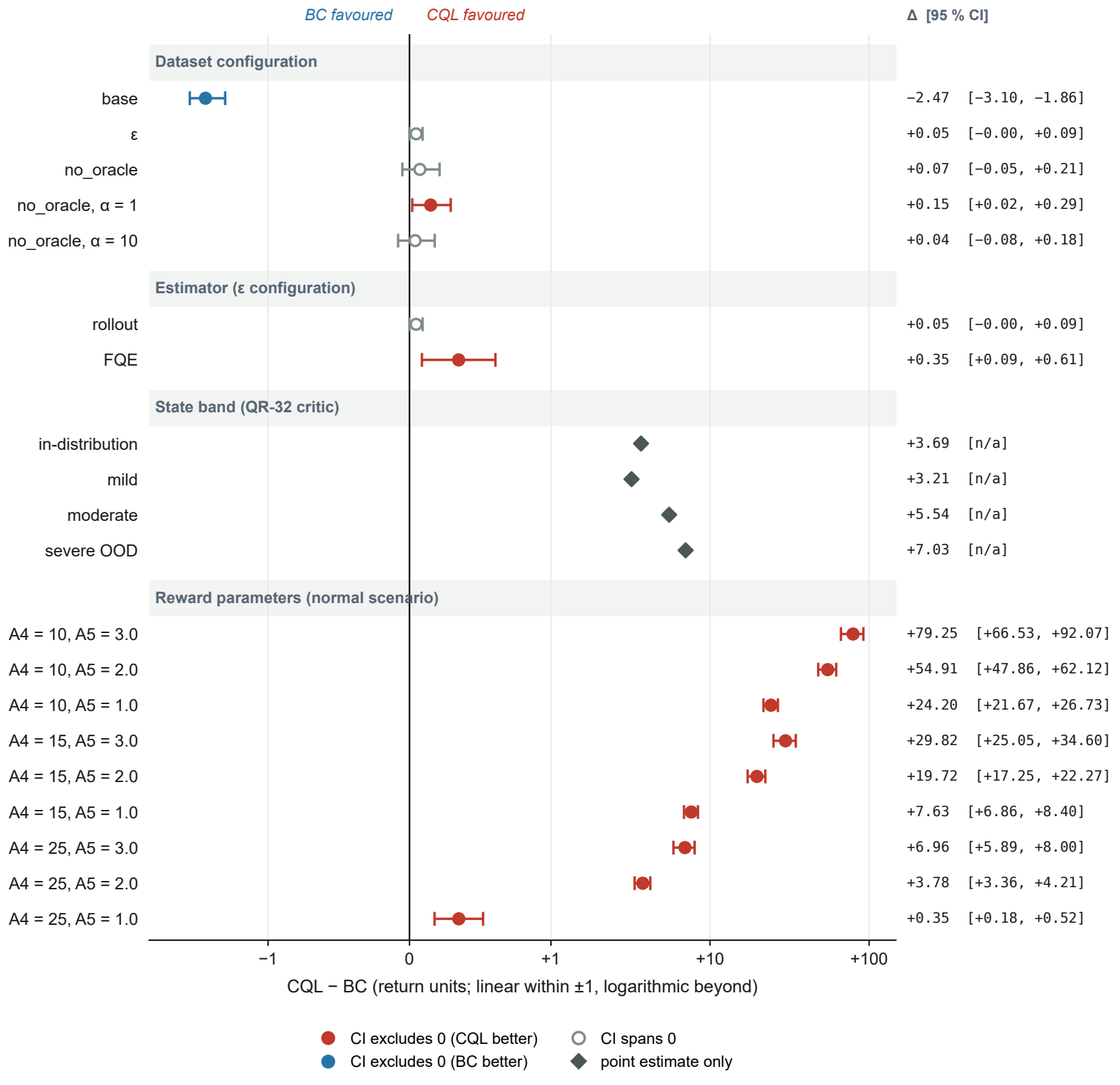

BC favoured
CQL favoured
Δ [95 % CI]
Dataset configuration
base
−2.47 [−3.10, −1.86]
ε
+0.05 [−0.00, +0.09]
no_oracle
+0.07 [−0.05, +0.21]
no_oracle, α = 1
+0.15 [+0.02, +0.29]
no_oracle, α = 10
+0.04 [−0.08, +0.18]
Estimator (ε configuration)
rollout
+0.05 [−0.00, +0.09]
FQE
+0.35 [+0.09, +0.61]
State band (QR-32 critic)
in-distribution
+3.69 [n/a]
mild
+3.21 [n/a]
moderate
+5.54 [n/a]
severe OOD
+7.03 [n/a]
Reward parameters (normal scenario)
A4 = 10, A5 = 3.0
+79.25 [+66.53, +92.07]
A4 = 10, A5 = 2.0
+54.91 [+47.86, +62.12]
A4 = 10, A5 = 1.0
+24.20 [+21.67, +26.73]
A4 = 15, A5 = 3.0
+29.82 [+25.05, +34.60]
A4 = 15, A5 = 2.0
+19.72 [+17.25, +22.27]
A4 = 15, A5 = 1.0
+7.63 [+6.86, +8.40]
A4 = 25, A5 = 3.0
+6.96 [+5.89, +8.00]
A4 = 25, A5 = 2.0
+3.78 [+3.36, +4.21]
A4 = 25, A5 = 1.0
+0.35 [+0.18, +0.52]
−1
0
+1
+10
+100
CQL − BC (return units; linear within ±1, logarithmic beyond)
CI excludes 0 (CQL better)
CI spans 0
CI excludes 0 (BC better)
point estimate only